

ECE298A

Integrated Circuit Design and Tapeout

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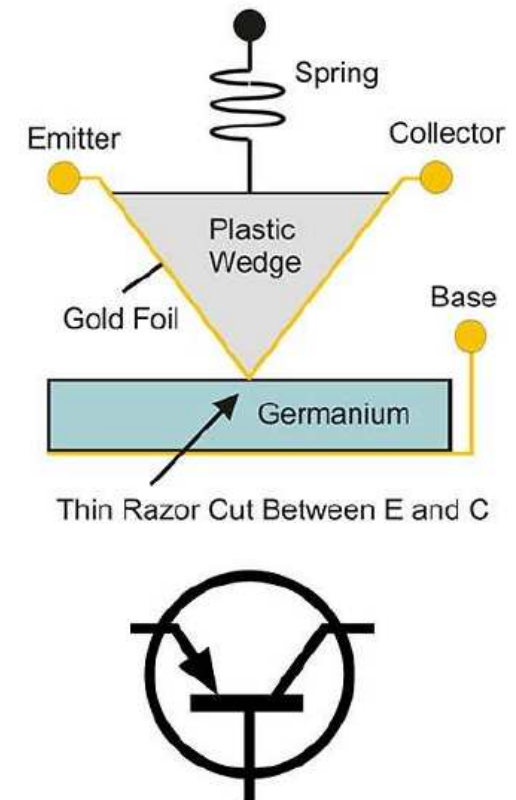
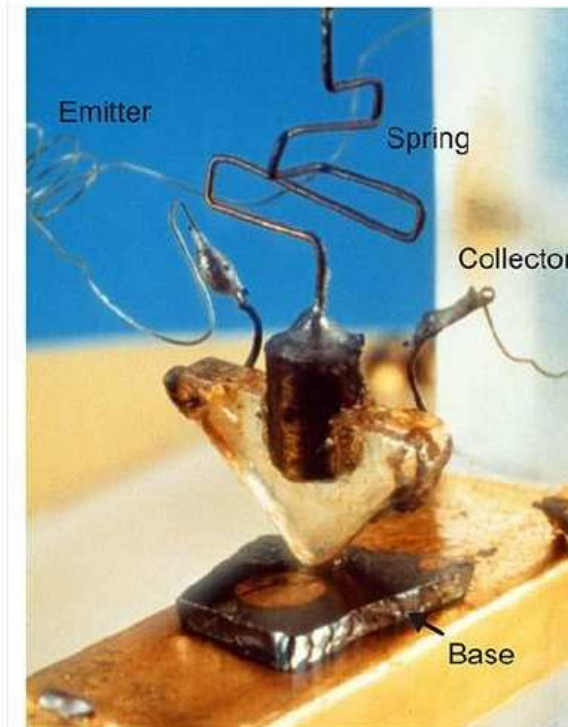
September 10, 2026

It Began Over 75 Years Ago

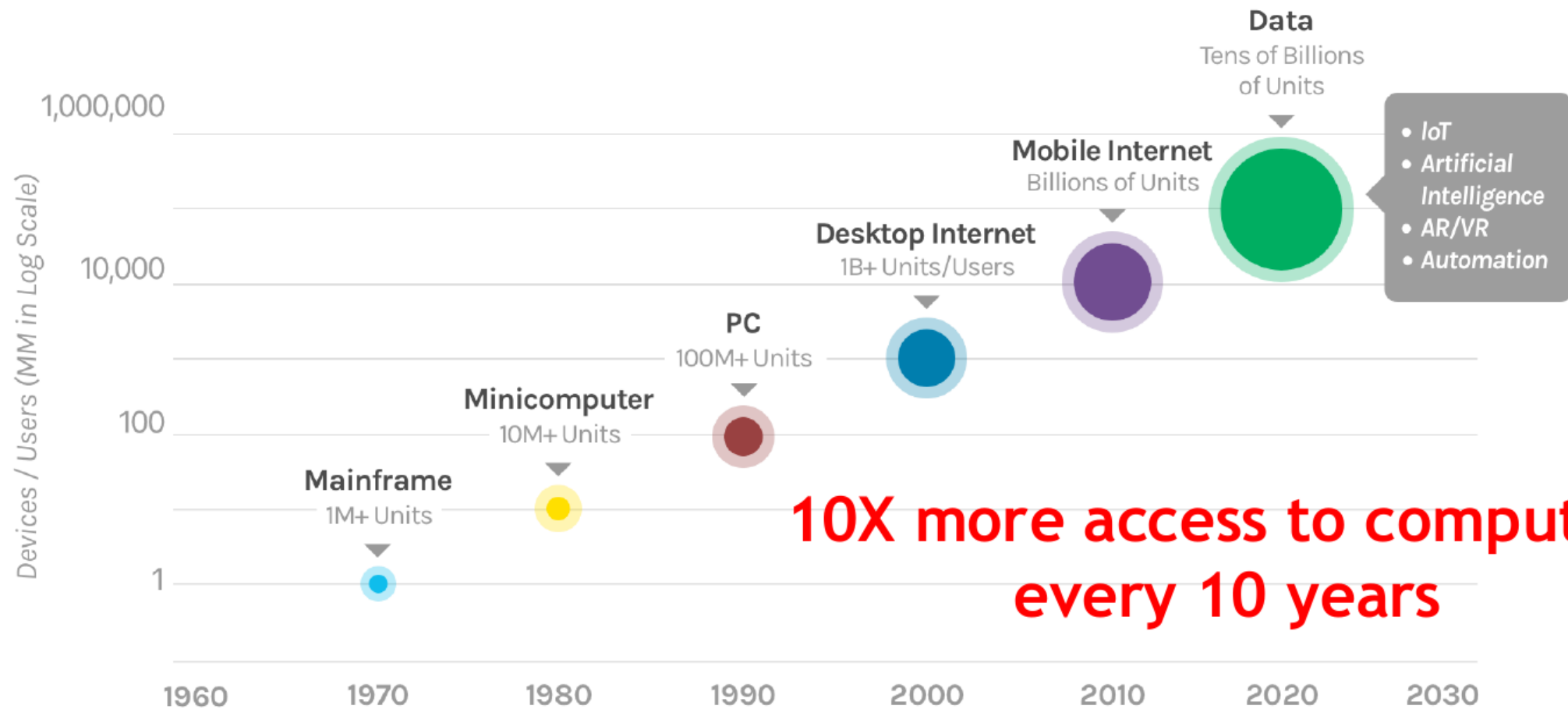


- First transistor was successfully demonstrated on December 23, 1947 at Bell Labs in Murray Hill, NJ (research arm of AT&T)
- Invented by John Bardeen, Walter Brattain & William Shockley

<https://www.nutsvolts.com/magazine/article/the-story-of-the-transistor>



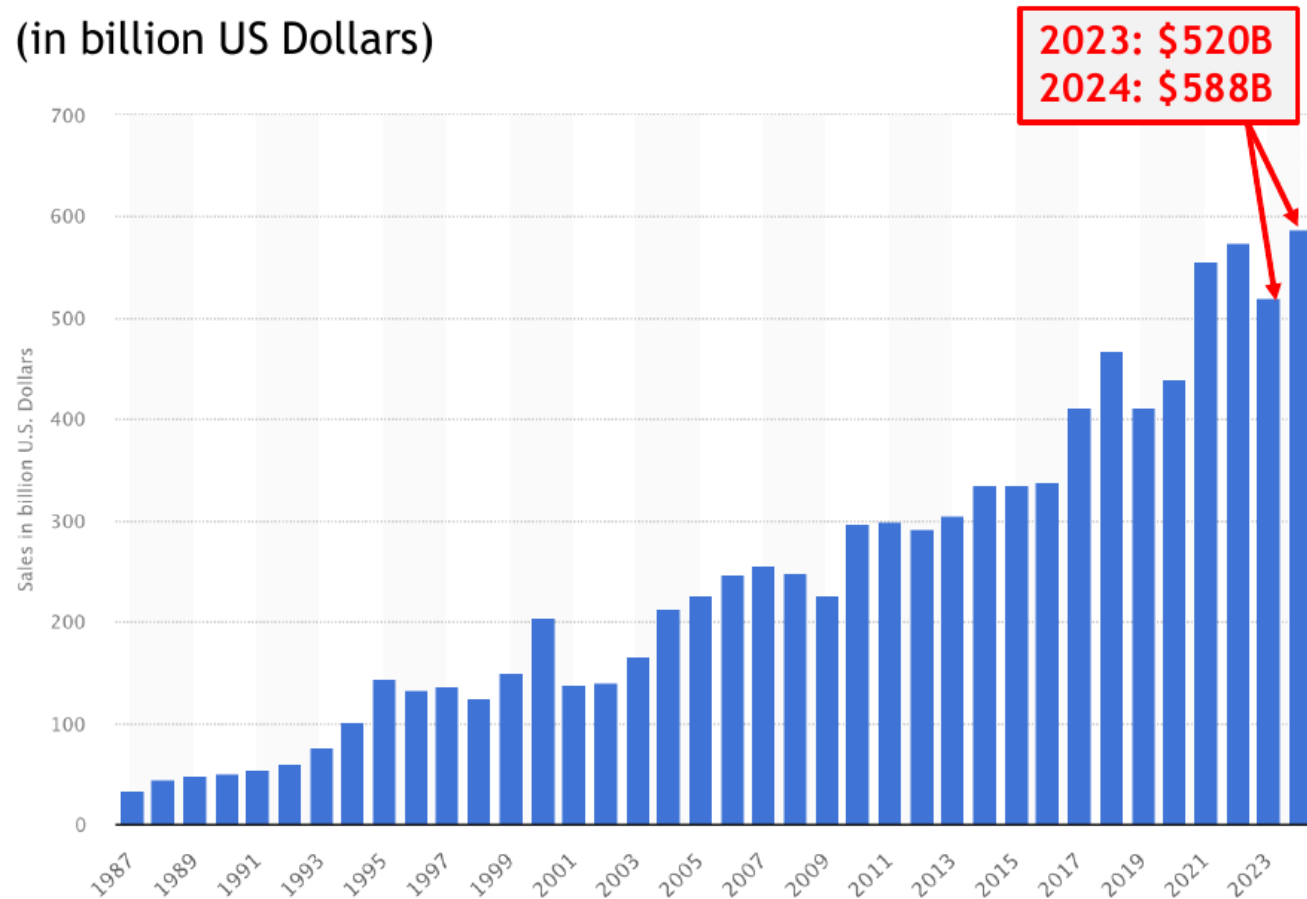
Semiconductor Demand



Source: Morgan Stanley
(Courtesy Alessandro Piovaccari)

Worldwide Market Size

(in billion US Dollars)



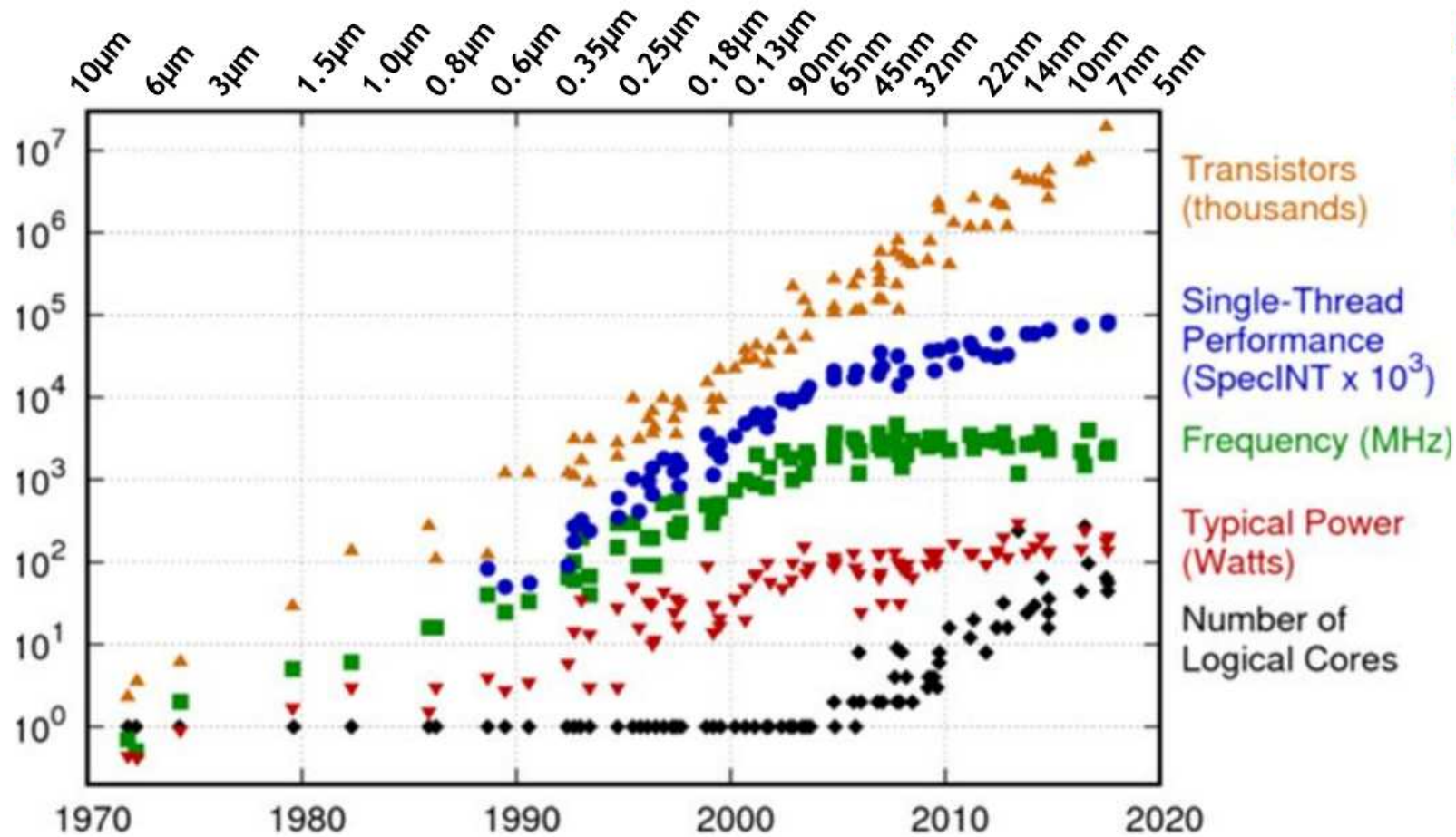
Let's put it in context...

→ Worldwide Markets (2021)

- Semiconductors: \$553B
- GDP: \$93,864B
- IT Data Centers: \$228B
- IT Devices: \$705B
- Car & Auto: \$3,600B
- Home Appliances: \$420B
- Semiconductor market expected to hit \$1T in 2030

Sources: Statista, IBIS World, SIA
Reference: [SIA2021], [Gartner2021]
(Courtesy Alessandro Piovacari)

What Makes All This Possible?



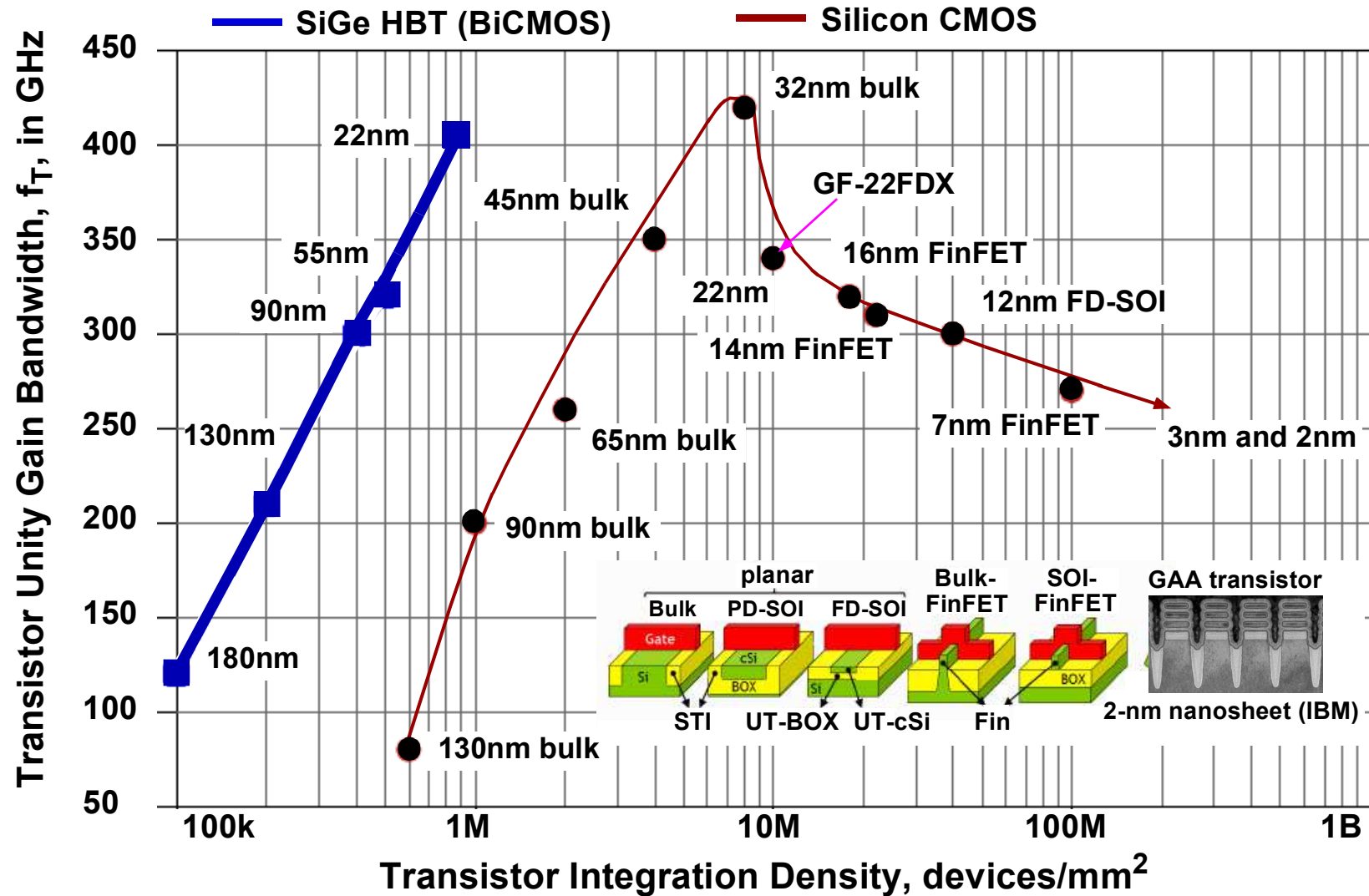
Source: <https://www.karlrupp.net/2018/02/42-years-of-microprocessor-trend-data/>

Miniaturizing (scaling) transistors makes them cheaper, faster, and more energy efficient

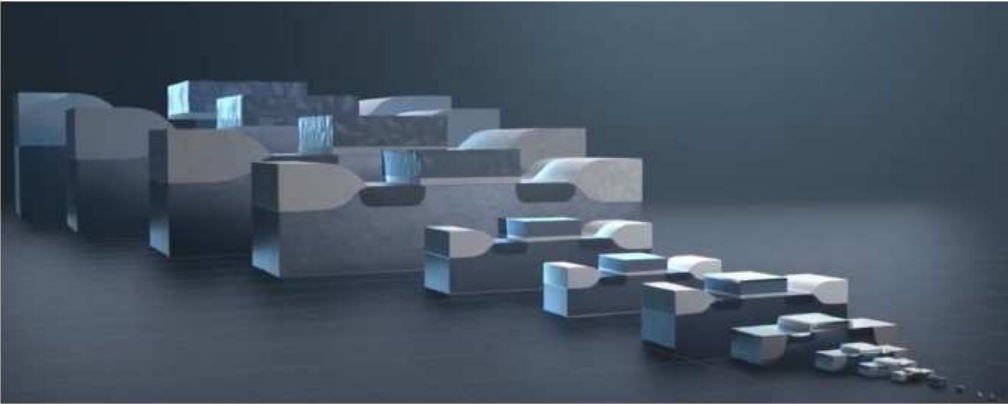


(Courtesy Alessandro Piovacari)

Silicon Technology Scaling



50 Years of Moore's Law



(Source: intel.com)

- 1,000,000X smaller
- 3,500X greater performance
- 60,000X lower cost
- 90,000X more energy efficient

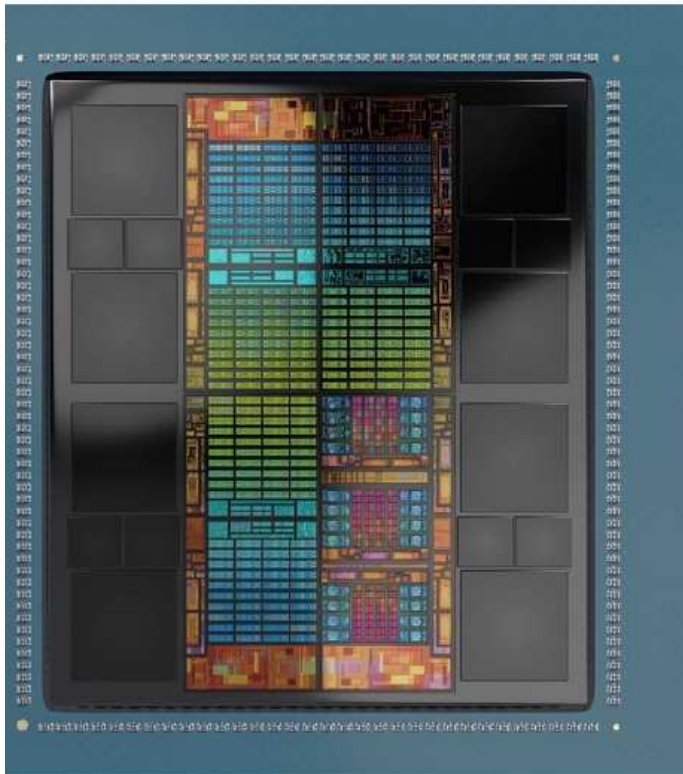
(Courtesy Joe DiFilippo)



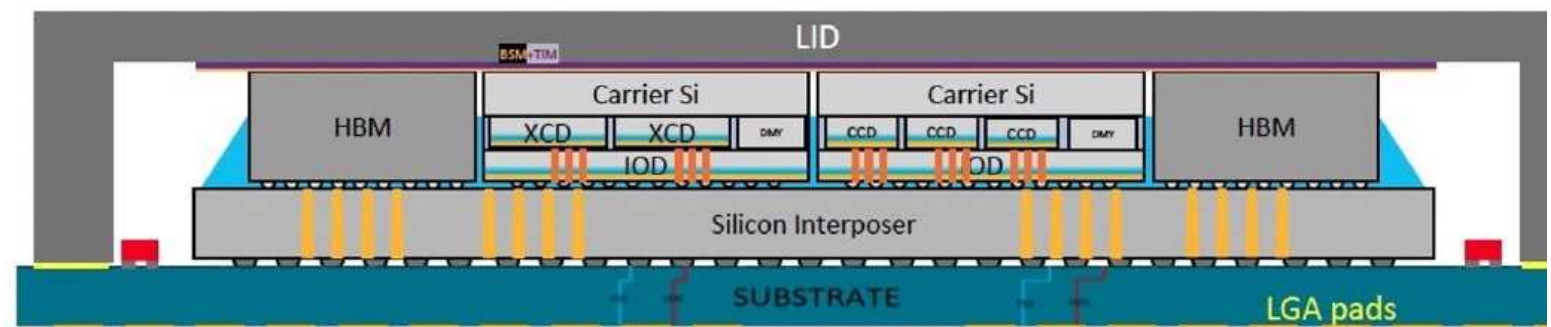
If car technology progressed at the same pace as semiconductors, the VW Beetle would:

- Go 300,000 mph
- Cost \$0.04
- Get 2,000,000 miles per gallon of gas
- Last your entire life on one single tank of gas

AMD MI300A Accelerated Processor Example

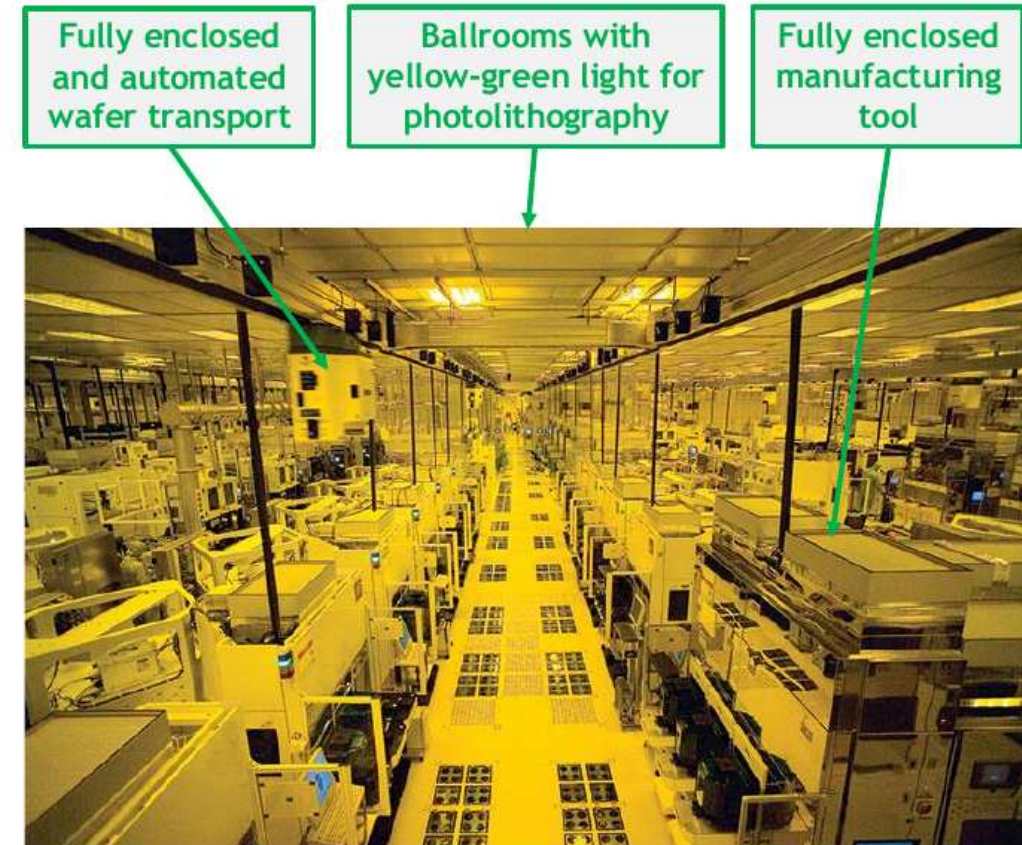


- SoIC 3D Technology
 - 3 CPU 5-nm chiplets (CCD)
 - 6 accelerator 5-nm chiplets (XCD) – AI, Matrix Multiplication
 - 4 Input-output 6-nm dies (IOD)
 - Silicon Carrier
- CoWoS
 - 3D computing system
 - HBM Memories
 - Silicon Interposer
- Total of 150B transistors



A Modern Mega-Fab (Wafer Fabrication Plant)

- Typical price tag of \$20B
- Throughput of 30-50k wafers per month
- Depreciates at \$0.5M per hour

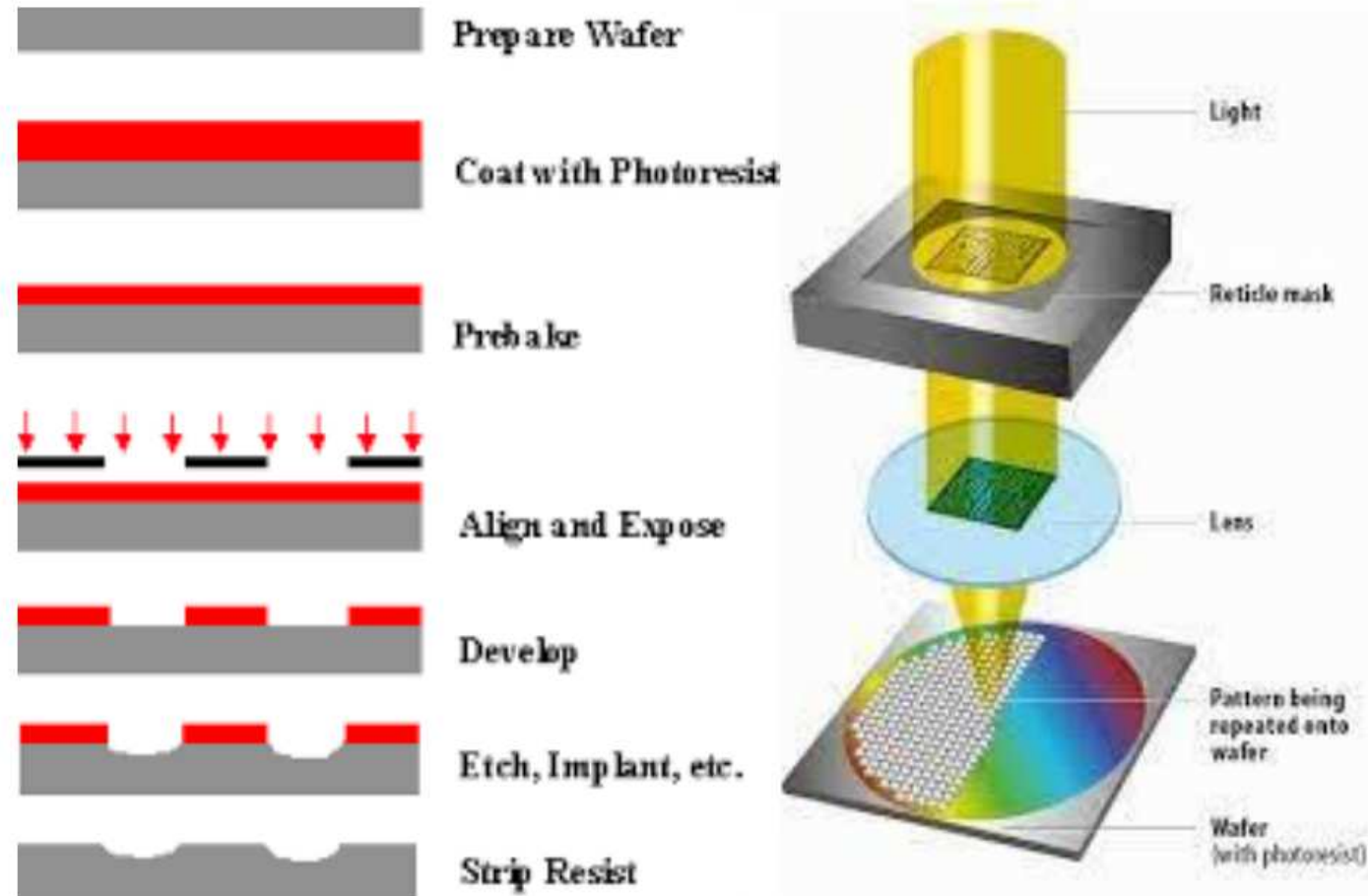


Source: WikiChip

(Courtesy Alessandro Piovaccari)

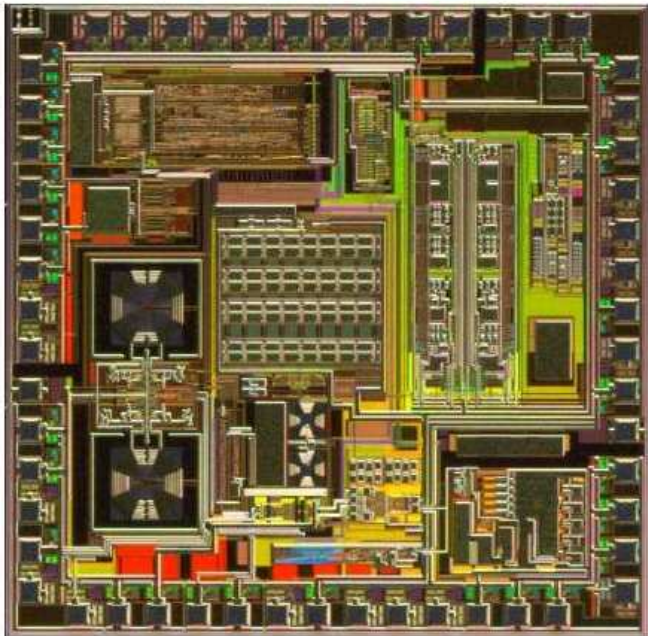
Same Effort Makes 1 or 1B Transistors

- Expose entire chip in one shot
- Pattern smaller features by using shorter wavelengths of light to expose photoresist (historically started at 436nm, now 13.5nm)
- Commonplace today to print nm-scale patterns
- Must be done in “clean room” free of dust particles that will destroy integrity of mask patterns



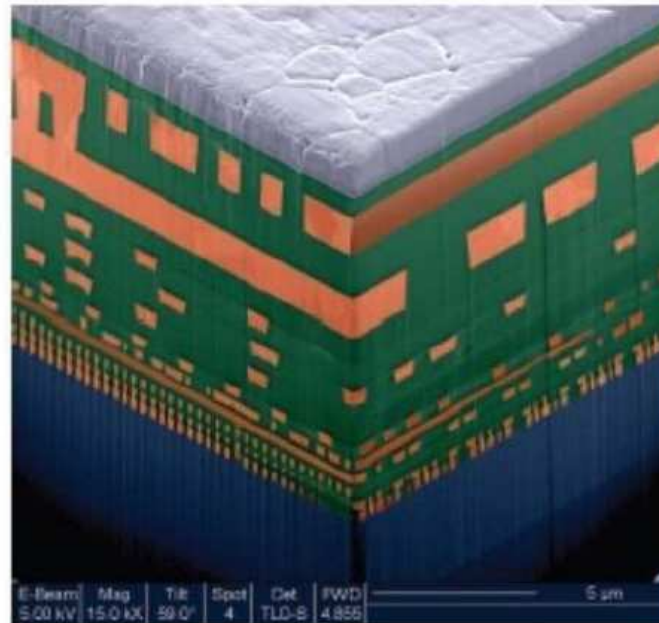
The Result

Top View of a Die



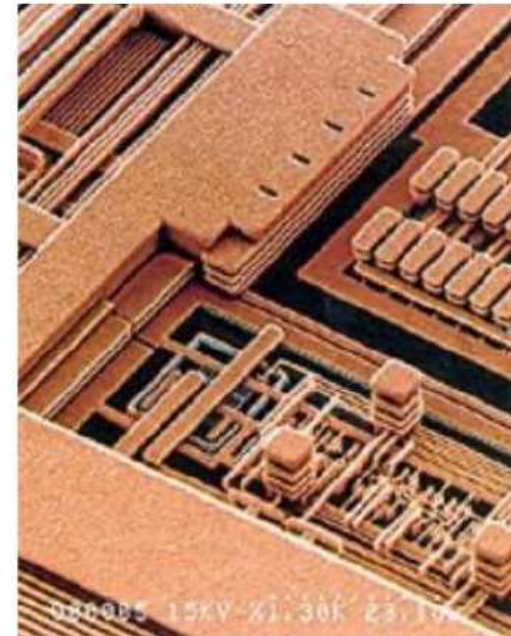
Source: IBM

Cross-section



Source: IBM

Metal layers

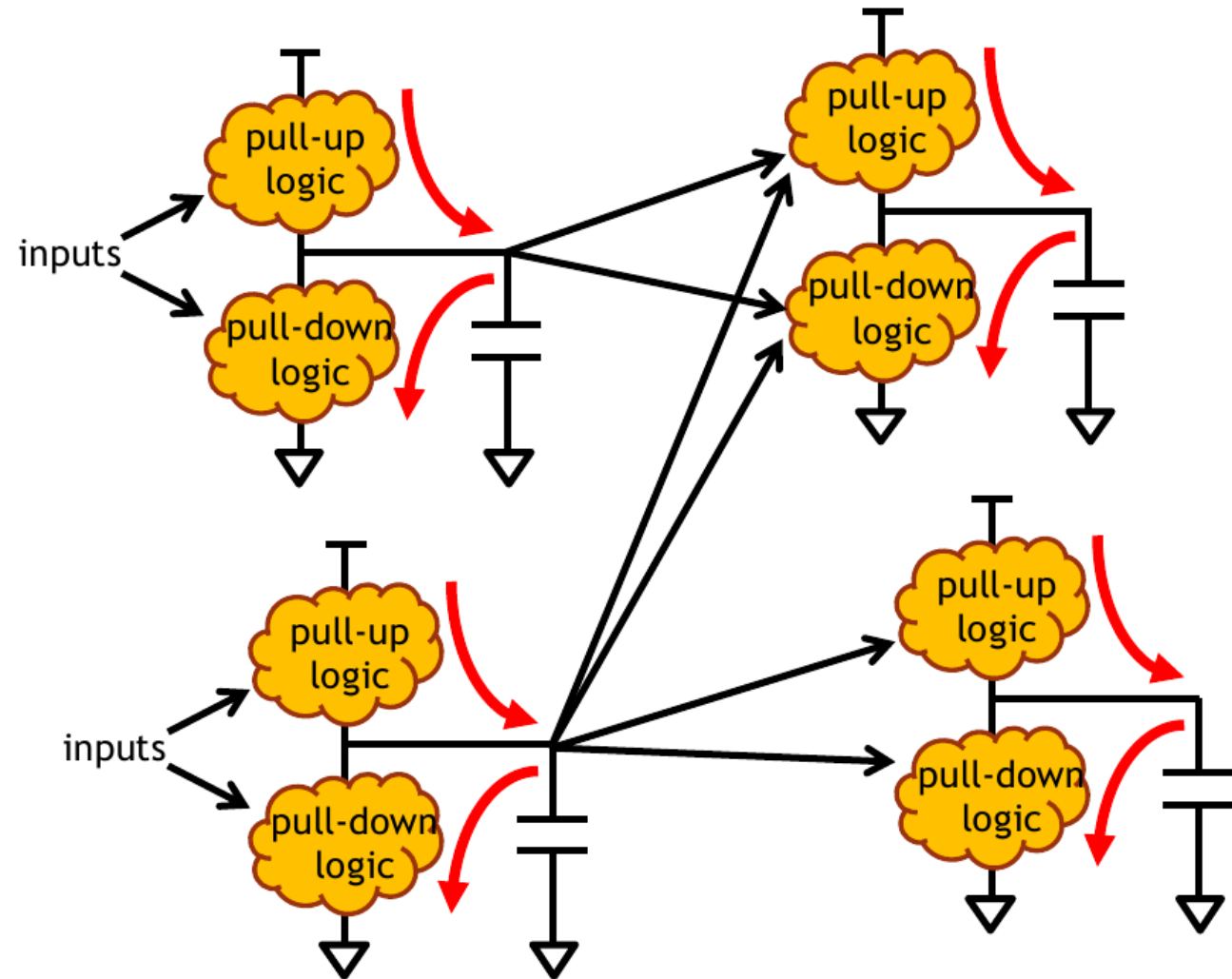


Source: IBM

(Courtesy Alessandro Piovaccari)

Circuits on a Chip

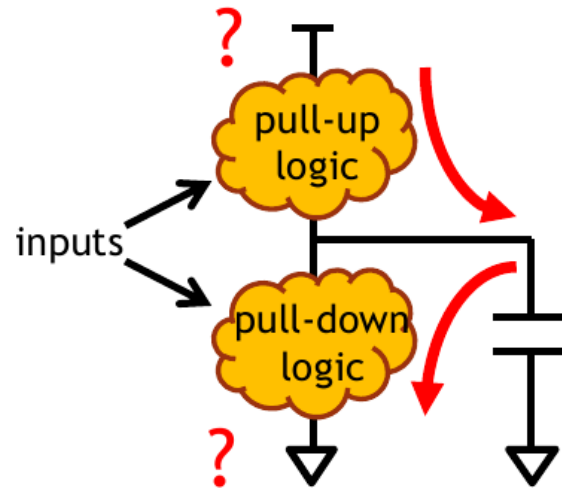
- Billions of independent little capacitors - voltage of each represents a state or stored information (e.g., logic 0 or 1)
- Capacitors charge/discharge (pull-up to supply or pull-down to ground) depending on logical function of inputs
- Logic consists of switches implemented using transistors
- Clocks control when logical states get updated



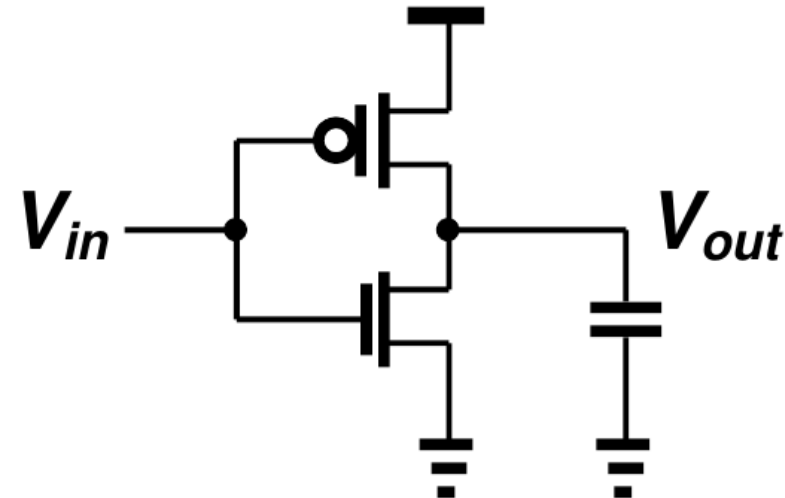
CMOS Inverter - A Simple Logic Gate



V_{in}	V_{out}
0	1
1	0



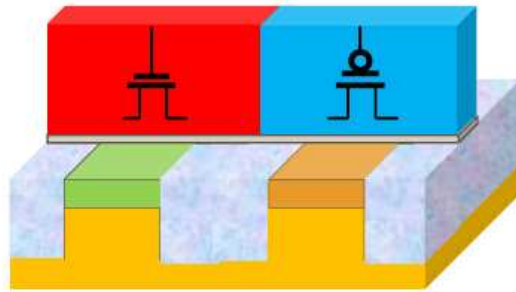
The Solution



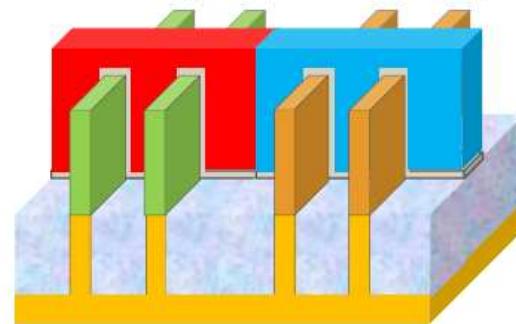
- $V_{in}=0 \rightarrow$ NMOS is off, PMOS is on $\rightarrow V_{out}=1$
- $V_{in}=1 \rightarrow$ NMOS is on, PMOS is off $\rightarrow V_{out}=0$
- No static current when input doesn't change $\rightarrow$ low power

Transistor Designs

Planar FET (10 μ m to 28nm)

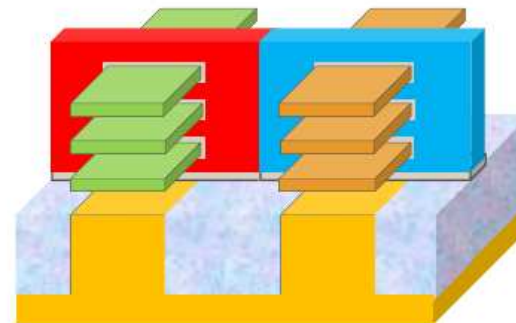


FinFET (22nm to 3nm)



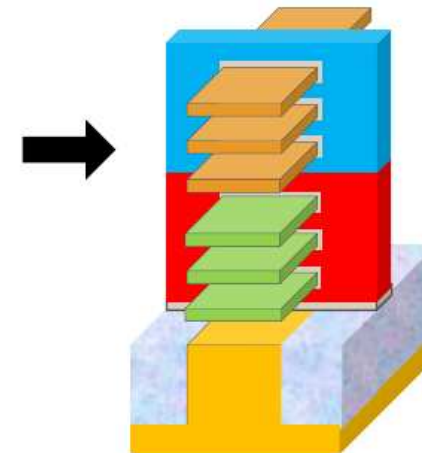
Latest and greatest we can buy today

Stacked Nanoribbons (2nm to 1nm)

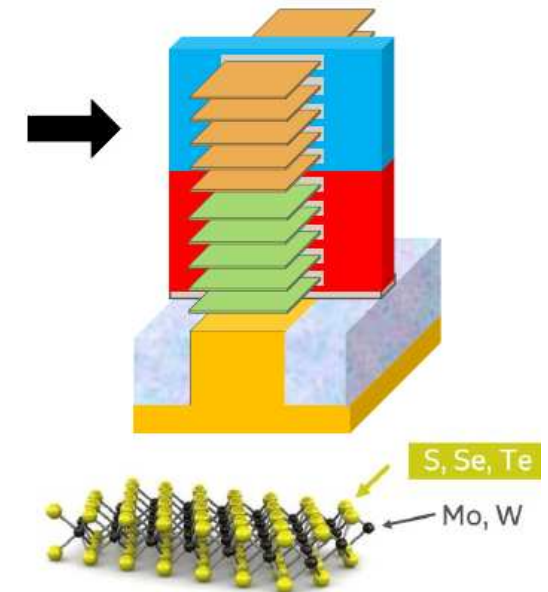


Latest and greatest we can buy in 2 years

Complementary FET (CFET) of Stacked Nanoribbons

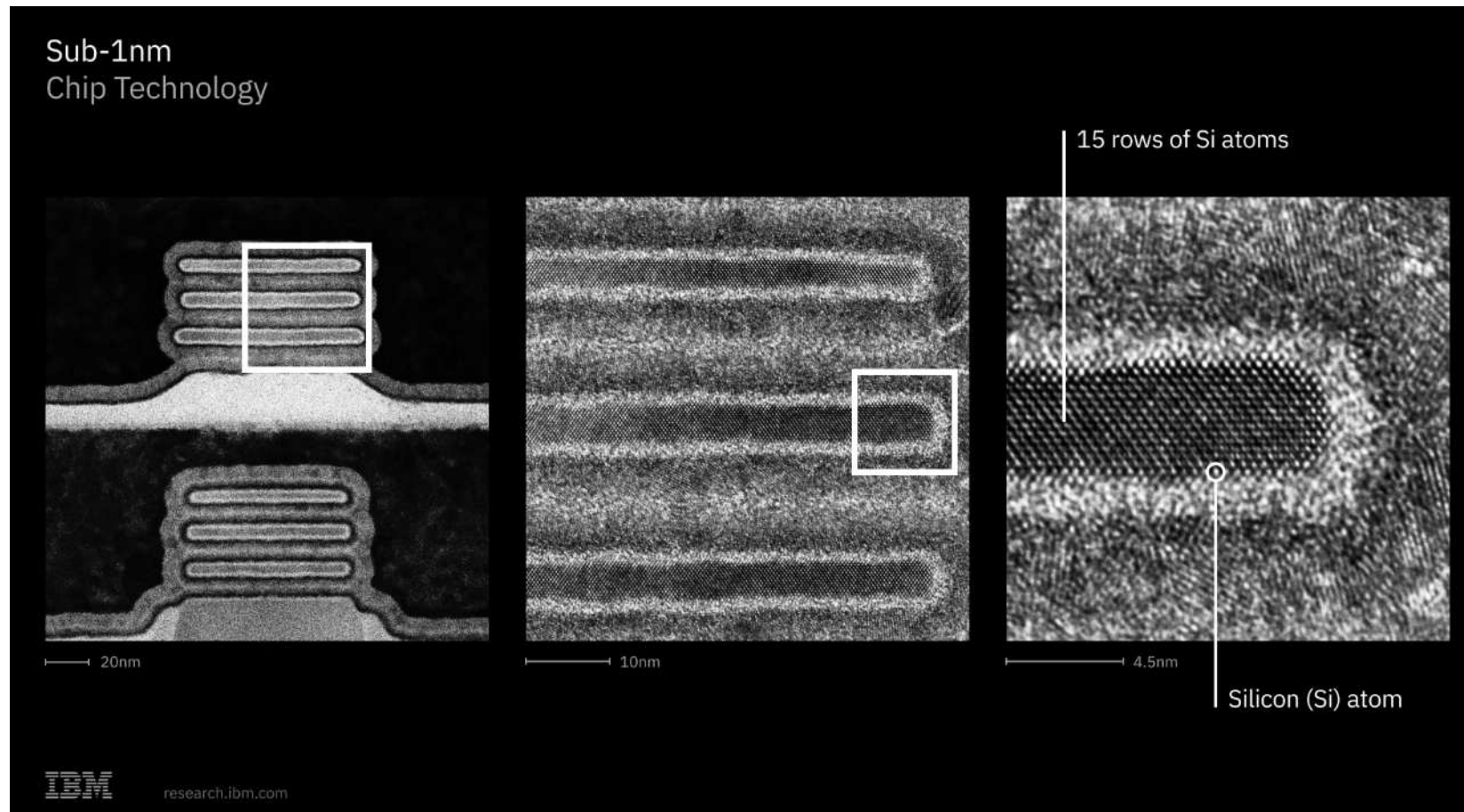


Complementary FET (CFET) of Stacked 2D Materials



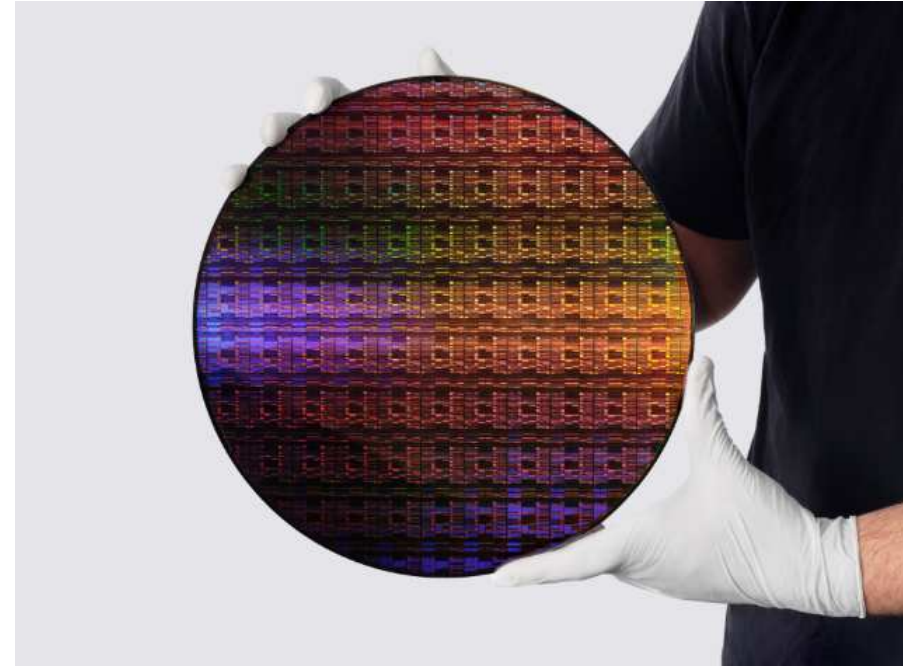
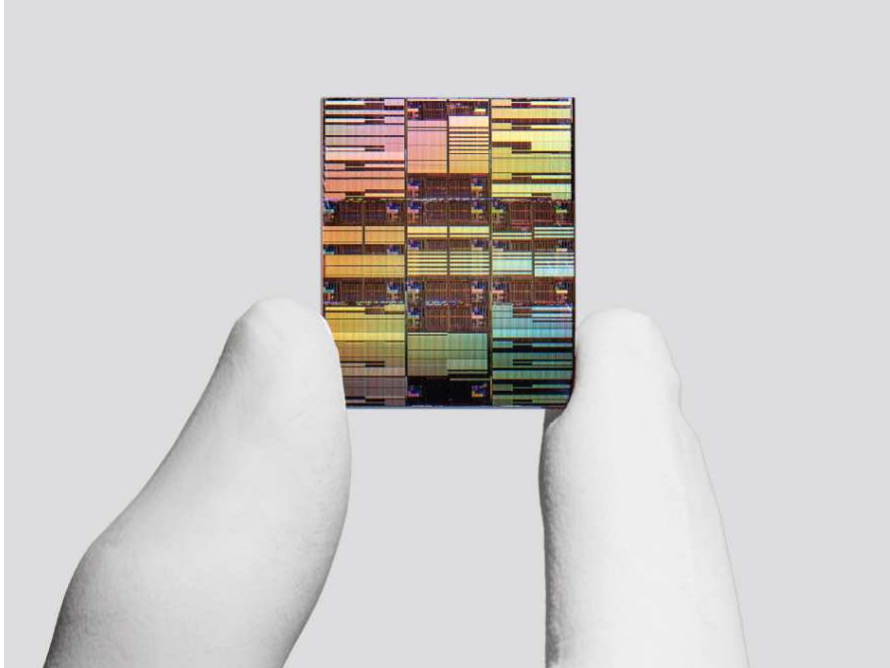
- Really complex
- Still building switches, just lots of them

IBM 7A Nanostack Transistors (2026)



- High Numerical Aperture Extreme Ultraviolet lithography developed by ASML enables creation of smaller, more powerful chips. IBM 7-Angstrom logic scales below the 1-nm node for the first time. Angstrom-level devices have dimensions approach the size of individual atoms

IBM 7A Silicon CMOS Demonstration



- Published results report up to 50% higher performance, or 70% greater energy efficiency than 2-nm chips
- IBM's new sub-1-nm chip packs ~100 billion transistors onto a chip the size of a fingernail, twice the density of 2-nm chips unveiled in 2021. IBM's semiconductor roadmap projects at least a decade of scaling into the future